

Certificates for Scoped Answers: Support, Alternative Exclusion, and Evidence Selection

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Abstract

Evidence can contain the right numbers and metadata while failing to establish which scope belongs to which result. We formalize a certificate for a complete joint answer: every required claim has positive source support, every modeled answer-changing rival is refuted, no active judgment conflicts, and an external coverage assessment accounts for admissible alternatives outside the proposal set. We prove source-relative soundness and conditional completeness, separate joint error from selective risk, and show precisely where missing candidates and fallible verification invalidate assurance. A finite optimization model represents conjunctive witnesses, shared source spans and adverse judgments. We give exact formulations, a classical set-cover reduction, a counterexample to fixed-lookahead guarantees, and a restricted cover approximation with explicit assumptions. A reference checker and exhaustive solver accompany the theory. These are conditional results and authored implementation checks; no natural-data improvement or general semantic-verification guarantee is established. The intended contribution is a testable certificate-based method for scoped retrieval, whose empirical advantage remains to be measured.

1 Problem and contribution boundary

A source may report one quantity for a segment in thousands and another for a consolidated population in millions. A continued table row, a unit header and a population phrase can all be present without the relations that connect them. For a question comparing two document versions, both requested versions belong to the same joint answer. The retrieval task is to obtain the evidence needed to warrant that joint answer and distinguish relevant alternative bindings. The source corpus, interpretation rules and output contract must be fixed; “the newest document wins” is not a general resolution rule.

The starting point is established work. Distinguishing decision-equivalent hypotheses is the purpose of equivalence-class determination. Sufficient-context, gap-judging and complementary-evidence methods already address important parts of evidence adequacy. Conjunctive sufficient evidence and set-level retrieval also appear in recent work. Therefore neither hypothesis partitioning nor the statement that evidence must be jointly sufficient is claimed as new here. Section 7 specifies the overlap and the remaining empirical test.

This manuscript closes the mathematical contract for the proposed route. Its contributions are a precise integration of source support, rival exclusion, candidate coverage and complete joint scope; proofs of the resulting conditional guarantees; and an optimization and implementation contract that does not confuse known-map packing with online acquisition. The elementary reductions and cover bounds are attributed adaptations of standard machinery. The source-semantic assumptions are exposed rather than supplied by the finite checker. No claim of a fundamentally new optimization family or an ACL-ready empirical paper is made.

2 A finite certificate for complete scoped answers

The following results specify a certificate contract. They assume source semantics, query interpretation and coverage properties that the supplied-map selector does not itself establish. In particular, exact checking of a finite certificate is not a proof that a learned verifier is sound or that the candidate set is complete.

Joint actions. Fix a question q , an immutable admitted corpus D , and a declared policy P for query readings, source authority, version identity, normalization and permitted inference. Let R_q be the finite ordered set of required output slots. A result slot contains its value, concept, entity, period, unit/scale, population, analysis definition and source-version identity. A comparison slot also identifies its endpoint slots, relation type and material qualifiers. Reporting two values, computing their difference, asserting a comparable change and asserting an explicit correction are distinct actions. Comparison claims require their own source premises or a registered checked derivation.

A hypothesis h proposes a *complete joint answer* for R_q , together with a nonempty finite set of necessary source claims $C(h)$ and their binding or derivation structure. Every material output claim and attachment must occur in $C(h)$ or be derived explicitly from it. Let $K_q(h)$ state structural completeness and faithful eligibility under the explicit question constraints. The universe $\mathcal{U}$ contains eligible anchored joint interpretations; the proposal set $H \subseteq \mathcal{U}$ is finite and nonempty. The universe need not be finite or mechanically enumerable.

The canonical answer map $a_q(h)$ preserves every requested version and comparison relation. Define

$$h \sim_q g \iff a_q(h) = a_q(g). \quad (1)$$

This is an equivalence relation by equality of canonical records. Different parse trees or supporting citations may be equivalent if they induce the same scoped action. Equal numbers do not erase different entities, populations, periods or source versions. For example, a question requesting original and amended values has a two-version joint answer even when the values agree; those versions are not rival hypotheses to be eliminated. A supplied symmetric incompatibility graph is insufficient unless it faithfully separates distinct actions. In particular, a nontransitive “compatible enough” relation does not define the quotient in (1).

Source consistency and warrant. Let $\mathcal{M}(D)$ be a nonempty class of models satisfying the admitted source assertions under P . Provenance-scoped assertions such as “version one reports 12” and “version two reports 15” can both hold. Genuine source conflict must be represented explicitly or excluded from this coherent-model theorem; classical inconsistency cannot be used to entail arbitrary answers. For $\phi_h = \bigwedge_{c \in C(h)} c$, define

$$T = \{h \in \mathcal{U} : \exists m \in \mathcal{M}(D), m \models \phi_h\}, \quad (2)$$

$$W = \{h \in \mathcal{U} : D \models \phi_h\}. \quad (3)$$

We consider resolution tasks with $T \neq \emptyset$. Since $\mathcal{M}(D) \neq \emptyset$, $W \subseteq T$. Thus T contains eligible source-consistent joint interpretations, while W contains proposals whose required claims are all positively warranted. There may be multiple admissible actions in T , including different readings of an underspecified question.

Source warrant and unique query resolution are deliberately separate. If “positive support” were defined to mean that an output is already the uniquely correct full answer under every admissible reading, support alone would establish correctness and the coverage theorem would be redundant. Our support relation warrants the claims of a complete eligible proposal; the certificate additionally establishes uniqueness over T . Assurance is relative to D, q, P , not to undeclared sources, source authenticity, real-world truth, or first-public availability.

2.1 Conjunctive witness semantics

A source span identifies an immutable source, exact offsets or coordinates, and text identity. A witness rule w names a candidate h , a necessary claim $c \in C(h)$, a sign $\sigma \in \{+, -\}$, a nonempty finite dependency set D_w of source spans, and a derivation/provenance record. If $\text{cov}(E)$ is the union of actually delivered spans, activation requires

$$w \text{ active in } E \iff \text{every span in } D_w \text{ is covered by } \text{cov}(E). \quad (4)$$

All leaf dependencies of a multi-step derivation must be included. Bundle names, nearby tokens, and generated explanations do not replace missing premises.

A rule is semantically sound when its anchored assertions jointly entail c for sign $+$, or $\neg c$ for sign $-$, under P . Faithful extraction then gives $D \models c$ or $D \models \neg c$, respectively. Presence of valid offsets and hashes checks source identity and coverage, not that entailment.

Write $P_E(h, c)$ and $N_E(h, c)$ for the existence of an active positive and negative rule. The claim is supported when $P_E \wedge \neg N_E$, contradicted when $N_E \wedge \neg P_E$, and unresolved otherwise. Record a separate conflict flag for $P_E \wedge N_E$. Missing evidence, retrieval failure and lack of proof remain unresolved; they never imply contradiction. Define

$$\text{Pos}_E(h) \iff \text{all } c \in C(h) \text{ are supported}, \quad (5)$$

$$\text{Ref}_E(h) \iff \text{some } c \in C(h) \text{ is contradicted}, \quad (6)$$

$$V_E = \{h \in H : \neg \text{Ref}_E(h)\}. \quad (7)$$

The conservative certificate rejects any active claim conflict, including one on a rival candidate. Harmless same-action survivors need not all be supported.

Lemma 2.1 (Local semantic soundness). *If all active rules are sound and $\mathcal{M}(D)$ is nonempty, no active claim has a positive/negative conflict. Further, $\text{Pos}_E(h)$ implies $h \in W$, and $\text{Ref}_E(h)$ implies $h \notin T$.*

Proof. A conflict would entail both c and $\neg c$ in every member of a nonempty $\mathcal{M}(D)$, which is impossible. If each necessary claim is supported, every model satisfies every conjunct of ϕ_h , so $D \models \phi_h$. If one necessary claim c is contradicted, every model satisfies $\neg c$ and none can satisfy ϕ_h , which contains c . $\square$

2.2 Coverage and certificate soundness

Definition 2.2 (Valid answer-class coverage). The candidate set has coverage, denoted $\text{COV}(H; D, q)$, when

$$\forall t \in T \exists h \in H \cap T : a_q(h) = a_q(t). \quad (8)$$

This requires a *source-consistent* representative for every admissible answer class. Merely placing every action label in $a_q(H)$ is insufficient: an invalid representative can be soundly refuted while a valid omitted interpretation with the same action remains. Coverage of one reference interpretation is also weaker than (8) for an ambiguous task.

An explicit unknown sentinel begins OPEN. An operational decision $\widehat{\text{COV}}$ sets it to COVERED only through a declared coverage assessment. Soundness of this assessment means $\widehat{\text{COV}} \Rightarrow \text{COV}$. Beam exhaustion, low pair loss, confidence, or removal of all proposed rivals is not a coverage proof. A finite formal grammar may permit exhaustive domain-relative enumeration; a natural-language audit or estimator remains fallible and must be evaluated. The unknown sentinel is not an ordinary candidate that a finite negative witness can eliminate. Its justification is logically separate, not assumed statistically independent of other components.

Definition 2.3 (Supported unique-answer certificate). For exact declared evidence cost $\text{cost}(E)$ and budget B , a certificate chooses E and a winner $h^* \in H$ satisfying:

1. $\text{cost}(E) \leq B$;
2. there is no active positive/negative claim conflict;
3. $\text{Pos}_E(h^*)$;
4. every $g \in H$ with $a_q(g) \neq a_q(h^*)$ satisfies $\text{Ref}_E(g)$;
5. $\widehat{\text{COV}}$ holds.

The certificate records one complete support proof for h^* , an explicit refutation for every rival candidate, and coverage-assessment provenance.

Every member of each rival class must be refuted, unless a sound universal class-level argument covers them all. Refuting one representative does not refute a class. Positive support implies a nonempty survivor set; the gate is equivalently a fully supported survivor plus agreement of all survivors, with conflict, coverage and cost conditions retained. A zero weighted objective does not replace this logical condition, particularly with zero-weight rivals or an empty survivor set.

Theorem 2.4 (Source-relative certificate soundness). *Assume faithful query/schema and answer mapping, faithful source extraction and output rendering, a nonempty coherent $\mathcal{M}(D)$, sound active witness rules, and a sound coverage sentinel. A certificate as in Definition 2.3 establishes $h^* \in W$ and*

$$\{a_q(t) : t \in T\} = \{a_q(h^*)\}. \quad (9)$$

It therefore authorizes a complete positively warranted joint answer with a unique action in the admitted source/query domain.

Proof. Positive support and Lemma 2.1 imply $h^* \in W \subseteq T$; the left side of (9) thus contains the winning action. For any $t \in T$, sound coverage supplies $h \in H \cap T$ with $a_q(h) = a_q(t)$. If this differed from $a_q(h^*)$, the rival-refutation gate would imply $\text{Ref}_E(h)$; Lemma 2.1 would then give $h \notin T$, a contradiction. Every admissible interpretation therefore has the winning action. Its complete support derivation warrants each required claim, faithful schema checking includes every requested slot, and faithful rendering adds no assertion. The budget condition establishes only the declared evidence-cost bound and plays no role in semantic entailment. $\square$

Corollary 2.5 (Reference-target preservation). *Fix $t^* \in T$. Replace full coverage in Theorem 2.4 by the existence of $h \in H \cap T$ with $a_q(h) = a_q(t^*)$. An accepted output remains positively warranted and equals $a_q(t^*)$, but the premise does not establish uniqueness over all of T .*

Proof. If the output differed from the valid representative of t^* , that candidate would be refuted, contradicting local soundness. Positive warrant follows as before. No premise supplies a representative for any other admissible class, so no conclusion about those classes follows. $\square$

2.3 Finite checking and conditional completeness

Let $\mathcal{A}$ be a finite set of admissible source-span bundles and let the rule list be finite. A selection $E \subseteq \mathcal{A}$ delivers the union of its spans, which can activate a rule across multiple bundles. Let $\mathcal{F}_B$ be the selections satisfying Definition 2.3 for some winner, with all supplied objects and the cost renderer fixed.

Proposition 2.6 (Exact finite-map checking). *Exhaustive evaluation of all $E \subseteq \mathcal{A}$ returns a certificate if and only if $\mathcal{F}_B \neq \emptyset$. If*

$$B^* = \min\{\text{cost}(E) : E \text{ satisfies gates 2-5}\}, \quad (10)$$

with $B^ = \infty$ for an empty set, a finite-budget certificate exists exactly when $B \geq B^*$.*

Proof. The finite power-set enumeration includes every permitted selection. Each selection is checked against the defining predicates of $\mathcal{F}_B$, so a returned member is feasible and every feasible member is eventually checked. When the set defining B^* is nonempty, finiteness ensures the minimum is attained. The budget equivalence follows from this definition, without any assumption that cost is monotone under adding bundles. $\square$

This establishes exactness on supplied maps, not semantic truth. A heuristic failure does not prove infeasibility. Minimizing pair loss while allowing abstention does not solve the certificate-existence problem.

Theorem 2.7 (Conditional semantic completeness). *Assume the soundness premises of Theorem 2.4, and suppose $T/\sim_q$ has one class with action a^* . Further suppose:*

1. *the justified coverage sentinel is COVERED;*
2. *some $h^* \in H$ with action a^* has an expressible sound positive witness for every claim in $C(h^*)$;*
3. *every $g \in H$ with action different from a^* has an expressible sound negative witness for a necessary claim;*
4. *a single admitted selection E_0 includes all these witness dependencies and satisfies $\text{cost}(E_0) \leq B$;*
5. *search is complete over the admitted finite selections.*

Then the procedure returns a certificate with action a^ .*

Proof. The joint dependency condition activates the selected positive and negative witnesses. Coherent global rule soundness excludes opposing active judgments by Lemma 2.1. Thus E_0 supports all claims of h^* , refutes every different-action candidate, is conflict-free and fits the budget. The covered sentinel supplies the remaining gate. Hence $\mathcal{F}_B \neq \emptyset$; complete search and Proposition 2.6 return a certificate. Theorem 2.4 gives its unique action a^* . $\square$

Semantic uniqueness alone implies none of the witness-expressibility, retrieval, verifier-recognition, joint-budget, coverage-assessment or search-completeness premises. Individually affordable witnesses need not fit together. This theorem therefore provides no completeness guarantee for an unrestricted language model or the greedy supplied-map selector.

A predeclared projection π onto a subset of requested slots gives an analogous theorem with answer map $\pi \circ a_q$ and complete support for the emitted projected claims. Alternatives differing only on withheld slots need not be separated. The resulting certificate is a partial answer and does not establish the omitted version or comparison relation.

2.4 Failure localization and selective risk

Consider a distribution over tasks and full adaptive runs. Let A denote emission of a complete answer and let Bad denote failure of the grounded unique-resolution conclusion. Retain the nonempty coherent source-model and eligible-domain setup. Initially assume exact query/action representation, source extraction, schema and rendering. Require every emitted run to pass the finite mechanical gate of Definition 2.3 and retain the selected support and refutation proofs. This protocol premise does not

assume that the supplied semantic judgments or coverage assertion are true; their failures are the error events below. On emitted runs, let E_C be a cleared but false coverage assertion, E_+ a false positive judgment in the selected complete support proof, and E_- a false negative judgment used to eliminate a source-consistent rival. Here “false negative” means erroneous contradiction, not failure to recognize support; reporting should use the clearer terms *false support* and *false elimination*.

Theorem 2.8 (Error localization and risk bound). *Under those exact-component and mechanically gated emission assumptions, without assuming correctness of the supplied semantic judgments or coverage assertion,*

$$A \cap \text{Bad} \subseteq E_C \cup E_+ \cup E_-. \quad (11)$$

With $E_V = E_+ \cup E_-$,

$$\begin{aligned} \Pr(A \cap \text{Bad}) &\leq \Pr(A \cap E_C) + \Pr(A \cap E_C^c \cap E_V) \\ &\leq \Pr(E_C) + \Pr(E_+) + \Pr(E_-). \end{aligned} \quad (12)$$

If $p = \Pr(A) > 0$, selective risk satisfies

$$\Pr(\text{Bad} \mid A) \leq \min \left\{ 1, \frac{\Pr(E_C) + \Pr(E_+) + \Pr(E_-)}{p} \right\}. \quad (13)$$

No independence assumptions are required.

Proof. Take an emitted run outside $E_C \cup E_+ \cup E_-$. Its selected complete positive proof is sound, so the winner is warranted. Correct coverage represents every source-consistent action by a source-consistent candidate. If any such action differed from the winner, the gate would have eliminated its valid representative, requiring a false elimination and contradicting the complement of E_- . Thus the answer is grounded and uniquely resolved, proving (11). Partition $A \cap \text{Bad}$ into the disjoint part in E_C and the part in E_C^c . By localization the latter lies in E_V . Taking probabilities and applying the union bound yields (12). Dividing the joint error by $\Pr(A) = p$ yields (13). $\square$

When $p = 0$, selective risk is undefined; zero joint error from always abstaining does not establish answer quality. Marginal omission/verifier rates cannot be reported as answered-risk bounds without the denominator or conditional calibration: answering on one percent of tasks and being wrong every time gives joint error 0.01 and selective risk 1.

If directly justified conditional quantities are available, put $\delta_C = \Pr(E_C \mid A)$ and $\delta_V = \Pr(E_V \mid A \cap E_C^c)$. A partition gives

$$\Pr(\text{Bad} \mid A) \leq \delta_C + (1 - \delta_C)\delta_V. \quad (14)$$

When the second conditioning event has zero probability its contribution is zero and no conditional value is needed. The theorem estimates none of these quantities; sampling, confidence and transfer assumptions remain empirical.

If query parsing, necessary-claim completeness, answer equivalence, extraction, concept normalization or final rendering can fail, introduce E_R for these errors. Localization becomes

$$A \cap \text{Bad} \subseteq E_R \cup E_C \cup E_+ \cup E_-. \quad (15)$$

The risk bound must gain the corresponding term. A two-term candidate/verifier bound is legitimate only after stating which other components are exact or included in the verifier event. For a fixed reference endpoint, Corollary 2.5 permits target-class coverage in place of full coverage, but does not certify absence of other admissible actions.

Adaptive calls. If at most K judgments are issued and, for every reachable prior history, the conditional probability that the next issued judgment is semantically wrong is at most η , then the probability of any erroneous call is at most $\min(1, K\eta)$. To see this, condition each issued-error indicator on its prior history, bound its expectation by η , and apply the union bound. With a bounded predictable stopping rule the same argument gives $\eta\mathbb{E}[N]$ for the number N of issued calls. Static average per-call accuracy does not imply the required uniform conditional bound; adaptive acquisition can concentrate on high-error cases. Counting all issued calls is conservative and avoids conditioning only on favorable selected judgments.

2.5 Counterexamples and limits

Proposition 2.9 (Coverage cannot be inferred from a finite survivor set). *Without a sound coverage or equivalent uniqueness premise, a procedure restricted to a finite candidate/judgment transcript cannot guarantee unique resolution over all source-consistent completions of that transcript.*

Proof. Take the transcript $H = \{h_a\}$ with a fully supported h_a and no modeled rival. In one admissible completion, $T = \{h_a\}$. In a second, $T = \{h_a, h_b\}$ with $a_q(h_b) \neq a_q(h_a)$; all supporting claims of h_a remain true. The unrepresented possibility was not examined, so the visible transcripts coincide. A deterministic procedure emits the same unique action in both, making a false uniqueness claim in the second. Condition a randomized procedure on its randomness and the same argument applies. If it emits with probability r on this transcript, its uniqueness claim fails with probability r in the second completion. Universal soundness requires abstention here or an additional premise that distinguishes the completions. $\square$

This does not make the supported facts of h_a false. A perfect oracle for full task correctness already solves factual correctness; the impossibility concerns additional uniqueness with source-claim support. If support omits a necessary binding, a missing correct candidate can produce a wrong answer, but that construction also violates the complete-support contract.

Semantic judgments are essential. Let the source report 7 and let H contain full candidates for 7 and 9. Coverage is complete. False signed rules supporting 9 and refuting 7 can carry valid offsets and hashes, activate under every syntactic requirement and make the gate accept 9. Thus source identity cannot replace semantic soundness. Alternatively, a false elimination can remove one genuinely grounded query reading and produce false uniqueness while the selected winner’s source facts remain correct. Evaluating positive support alone cannot validate refutation behavior.

Why the remaining gates are required. A singleton survivor without evidence for its unit or population has no complete grounded support. Refuting every candidate also yields zero pair loss but no answer. A refuted member of a rival class leaves an unrefuted same-action member alive. A continuation-plus-header rule activated from the header alone has lost a necessary dependency. Two separately requested source versions cannot be collapsed into one action because their values agree. Each construction violates a different explicit gate or representation condition; none is repaired by increasing an objective weight.

Conflicts and monotonicity. With fallible supplied rules, adding evidence may activate positive support on a previously contradicted claim. Its state becomes unresolved with a conflict, the candidate reappears and authorization is blocked. Pair loss can increase; answerability is not unconditionally monotone. Under globally sound rules and coherent semantics, Lemma 2.1 excludes conflicts and negative conclusions persist as evidence is added, although budget feasibility need not persist.

The unproved components remain automatic query-faithful candidate generation, semantic source verification, natural-language coverage assessment, practical search, calibrated selective error and utility beyond strong structural and sufficiency controls at matched total costs. The certificate theory does not establish any of these empirical premises.

3 Finite certificate optimization

This section concerns a fixed, supplied witness map. Its exact optimum is a packing diagnostic once judgments are available; it is not the acquisition cost of discovering those judgments. The constructions are standard Boolean optimization and cover arguments integrated with the grounding contract. Their guarantees do not validate source semantics, candidate coverage, or unobserved verifier outcomes.

3.1 Actions, witnesses, and the optimization target

Let A be finitely many indivisible actions, E_0 the initially received evidence, and $E(X)$ the union of anchored intervals received from E_0 and actions $X \subseteq A$. Distinct immutable source versions never share an interval. Each hypothesis $h \in H$ has a nonempty required-claim set R_h and a complete joint answer action $a_q(h)$. Both versions of a requested comparison belong to this same hypothesis. Answer classes here are equality classes of normalized scoped actions; differing classes are incompatible.

A signed witness w concludes $(h, c, +)$ or $(h, c, -)$ from a nonempty set P_w of anchored source premises. Write

$$z_w(X) = \mathbf{1}\{P_w \text{ is covered by } E(X)\}, \quad p_{hc}(X) = \bigvee_{w:(h,c,+)} z_w(X), \quad n_{hc}(X) = \bigvee_{w:(h,c,-)} z_w(X).$$

All premises must be received; names of selected actions have no evidential meaning. Explicit unresolved judgments add neither sign. An active positive and negative judgment on the same claim is a conflict, wherever it occurs. For a chosen target h , the finite certificate constraints are

$$p_{hc}(X) = 1 \quad (c \in R_h), \quad (16)$$

$$\sum_{c \in R_g} n_{gc}(X) \geq 1 \quad (g \in H : a_q(g) \neq a_q(h)), \quad (17)$$

$$p_{gc}(X) + n_{gc}(X) \leq 1 \quad (g \in H, c \in R_g). \quad (18)$$

An independent cleared-coverage assessment is also necessary for authorization. Refutation in (17) concerns *every* hypothesis in each rival class, not one class representative. Same-action parses need not all have positive support. Zero-weight rivals are included.

For declared charged cost $C(X)$, minimize it over selections satisfying (16)–(18) for some h ; the minimum is $+\infty$ if none exists. Typed side constraints may separately limit acquisition cost and actual rendered input. Unknown coverage permits a *conditional* packing optimum, with authorization false. It cannot be cleared by optimization. A partial answer requires a separately declared projected task, rather than deleting an inconvenient claim of the joint task.

Proposition 3.1 (Equivalence to the finite answerability gate). *Under the answer-class contract, constraints (16)–(18), cleared coverage, and the budget constraints are equivalent to complete positive support for some survivor, a nonempty survivor set, no incompatible surviving pair, and no conflict or unresolved coverage blocker.*

Proof. The target is supported and cannot be contradicted because of (18); it therefore survives. Equation (17) excludes every different-action hypothesis, leaving compatible survivors. Conversely,

choose a completely supported survivor from an authorized selection. Every different-action hypothesis must have been excluded by a negative required claim; the other constraints are the remaining gates. These are finite algebraic statements, not proofs that the supplied judgments or coverage assessment are correct. $\square$

Pair loss may prioritize search but cannot replace these constraints: it can vanish after eliminating everyone, before positive support, or in the presence of zero-weight rivals.

3.2 An exact Boolean formulation with shared cost

Split each source at every action, initial-evidence, and premise endpoint, obtaining nonempty elementary intervals $e \in V$. Let A_e be actions providing atom e , b_e indicate initial coverage, and V_w be the premise atoms of witness w . Include unprovided gaps: an atom with $A_e = \emptyset$ cannot accidentally become received. This partition concerns coverage, *not* additive native token cost.

For binaries $x_a, u_e, z_w, p_{hc}, n_{hc}, t_h$, impose

$$\begin{aligned} u_e &= b_e \vee \bigvee_{a \in A_e} x_a, & z_w &= \bigwedge_{e \in V_w} u_e, \\ p_{hc} &= \bigvee_{w:(h,c,+)} z_w, & n_{hc} &= \bigvee_{w:(h,c,-)} z_w. \end{aligned} \tag{19}$$

An empty OR equals zero. Implement $y = \bigvee_{i \in I} v_i$ by $y \geq v_i$ and $y \leq \sum_i v_i$; implement $y = \bigwedge_{i \in I} v_i$ by $y \leq v_i$ and $y \geq \sum_i v_i - |I| + 1$. Then add

$$\begin{aligned} \sum_h t_h &= 1, & p_{hc} &\geq t_h & (h, c), \\ p_{hc} + n_{hc} &\leq 1 & & & (h, c), \\ \sum_{c \in R_g} n_{gc} &\geq t_h & & & (h, g : a_q(g) \neq a_q(h)). \end{aligned} \tag{20}$$

Both directions of each Boolean equivalence are essential: a solver must not hide an adverse activated witness by choosing $z_w = 0$.

For a nonnegative additive union-cost model, action a uses charge resources Q_a . Set $v_r = \bigvee_{a:r \in Q_a} x_a$, fixing already paid resources as appropriate, and minimize

$$C_0 + \sum_r c_r v_r + \sum_a k_a x_a, \quad c_r, k_a \geq 0. \tag{21}$$

A shared header resource is charged once; per-action cost k_a is separately charged. Resources and evidence atoms need not coincide. Different cost units require separate constraints or an explicit scalar conversion.

Proposition 3.2 (Exact finite formulation). *The binary program has exactly the evidence selections satisfying (16)–(18). With the declared additive resource contract, its objective is their exact charged cost.*

Proof. For an action selection, the first equivalence forces the actual source union; the ANDs force every witness’s actual activation; the remaining ORs force all positive and negative signs. Thus adverse judgments cannot be suppressed. A binary target $t_h = 1$ is possible precisely when all its claims are supported, every different-action rival is contradicted, and all claims are conflict-free. Conversely these conditions supply a feasible target and all forced variables. Resource ORs charge precisely the resource union, while the last term charges actions individually. The correspondence preserves the objective. Coverage remains an external gate. $\square$

If incompatibility is a general symmetric graph I , rather than the complement of an equivalence relation, introduce survivor binaries $s_h \leq 1 - n_{hc}$ for all c and $s_h \geq 1 - \sum_c n_{hc}$. Replace the rival constraints with $s_h + s_g \leq 1$ for $(h, g) \in I$ and $t_h \leq s_h$. Refuting only neighbors of the chosen target could otherwise leave an incompatible pair among *other* survivors.

3.3 Exact rendering and safe search

The existing renderer merges adjacent intervals, potentially removing source labels. Adding evidence can therefore reduce rendered cost; native tokenization also need not be additive across fragments. Equation (21) represents actual prompt tokens only with a proved renderer/tokenizer contract. Checking one incumbent does not establish that contract globally.

A generic exact alternative uses the renderer and tokenizer as a deterministic cost oracle. For each target, branch on action inclusion, maintaining selected S , excluded actions, and undecided U . Prune a branch if S already has a conflict or contradicts the target, or if even $S \cup U$ lacks a required target support or rival refutation. A conflict in $S \cup U$ alone is not a reason to prune. A cost prune requires a valid lower bound on *every* completion. Under arbitrary rendering, $C(S)$ need not be such a bound; zero is valid for a nonnegative oracle. Additive union costs permit $C(S)$ and linear-program relaxations as lower bounds. The same rule applies to each hard side budget.

Whenever S is a certificate, compare its exact cost with the incumbent. With nonmonotone cost, continue exploring descendants: a larger certificate may render more cheaply. Enumerate all unpruned leaves.

Proposition 3.3 (Exact oracle search). *Complete search with these pruning rules returns a minimum finite certificate for any deterministic nonnegative cost oracle.*

Proof. Every action subset is a leaf. Signed witness activation is monotone, so an existing conflict persists. An existing target contradiction either persists or becomes a prohibited conflict. If an obligation is absent from the union of all allowed remaining actions, no completion satisfies it. These prunes remove only infeasible completions. A valid cost lower bound removes only completions unable to improve the incumbent. Every other feasible leaf is compared with the exact objective; an optimum is retained or matched. The argument does not assume monotone claim states in the presence of conflicts. $\square$

Worst-case complexity is exponential. A time limit without an incumbent is unknown feasibility, not infeasibility. With an incumbent, report its exact cost and a valid global bound from unresolved branches or the exact MILP. A surrogate-cost gap is not a native-token gap after post-hoc retokenization. An arbitrary finite oracle can also be encoded by one variable per complete action-set configuration and a precomputed cost table, but that exact MILP has exponentially many configurations.

4 Limits of myopic selection

Proposition 4.1 (Classical set-cover embedding). *Minimum finite certificate cost is NP-hard even with singleton premises, nonoverlapping additive action costs, no conflicts, and supplied cleared coverage. The budget decision problem is NP-complete when costs are polynomially evaluable.*

Proof. Given weighted Set Cover with universe U , sets S_i , and costs c_i , create target h^* with one claim supported by anchored initial evidence, and a rival g_u for each $u \in U$. Rivals have different actions from h^* and no positive witnesses. Action i costs c_i and provides a unique span whose singleton witnesses contradict exactly the rivals g_u with $u \in S_i$. There are no negative witnesses for h^* or conflicts. Only h^* can be supported, so a selection is a certificate exactly when its sets cover U , at identical incremental

cost. A fixed initial charge can be added to the decision budget. A proposed certificate is polynomially checked by coverage, sign evaluation, and the stated cost algorithm, proving membership in NP. $\square$

This cost-preserving reduction inherits the classical Set Cover approximation barrier (Dinur and Steurer, 2014); it is not a new hardness theorem. A fixed base charge can be made negligible by scaling action costs. We do not identify the number of hypotheses, evidence atoms, and total input size when importing a logarithmic hardness parameter.

Proposition 4.2 (Conjunctive witnesses violate diminishing returns). *Refuted-rival count, obligation coverage, and fixed-weight pair-loss reduction need not be submodular, even with two-premise witnesses and no conflicts.*

Proof. Let h^* be initially supported and g an incompatible rival. Neither of two actions a, b produces a judgment alone, but together they activate a negative witness for g . For the refutation indicator f ,

$$f(\{b\}) - f(\emptyset) = 0 < f(\{a, b\}) - f(\{a\}) = 1.$$

The same instance has one refutation obligation and one positive-weight incompatible edge, giving the identical violation for the other objectives. $\square$

The deterministic single-realization case already violates the required marginal inequality, so adaptive submodularity is not automatic. EC2 analyzes a specified Bayesian test-outcome model (Golovin et al., 2010); arbitrary conjunctive source rules are not that model. Grouping complete premises into tests changes the ground set and its cost. Moreover, corrected Theorem 13 and its historical note in Golovin and Krause (2011), arXiv version 5, replace a flawed earlier general expected-cover argument. We import no EC2 constant and do not claim that EC2 itself has been disproved.

Define a k -lookahead improvement policy to admit only moves adding at most k primitive actions that immediately reduce pair loss, increase supported claims, or first enable answerability. Positive pair-loss density outranks zero density. This includes the proposed improvement-only singleton and two-bundle rules, but excludes deliberate zero-gain planning moves.

Proposition 4.3 (No finite ratio for fixed improvement lookahead). *For every fixed $k \geq 1$, this policy class has unbounded certificate-cost ratio with premises of arity $k + 1$, and can fail under a budget admitting a certificate.*

Proof. Initially h^* is completely supported and one incompatible rival remains. Actions $a_1, \dots, a_{k+1}$ cost one each; a negative witness for the rival requires all their spans. An alternative action d costs $M > k + 1$ and alone refutes it. No additional positive-support progress is possible. Every move of at most k cheap actions has zero gain; every improving move contains d and costs at least M . A maximum-density policy chooses d alone. The optimal certificate costs $k + 1$, so the ratio is at least $M/(k + 1)$, which diverges with M . A positive common initial charge preserves divergence. With budget $B = k + 1 < M$, there is no admitted improving move although all cheap actions form a feasible certificate. For two-bundle lookahead, three premises suffice. $\square$

This does not prohibit singleton actions in a globally planned algorithm. Such a planner can intentionally buy a zero-gain premise. It also does not deny that two-bundle lookahead repairs a particular two-part example.

5 A restricted classical cover guarantee

Fix a complete target h . Obligations O_h are its required positive claims and one refutation for *each* different-action hypothesis. Remove those already satisfied by E_0 and let $m = |O_h|$. Assume the following explicitly checkable finite-map conditions:

- (A) All eligible actions together activate no positive/negative conflict.
- (B) Each reachable useful witness has a unique inclusion-minimal activating action set D_w , so $z_w(X) = 1$ exactly when $D_w \subseteq X$ after conditioning on E_0 .
- (C) Incremental cost is the nonnegative union-resource cost $c(X) = \sum_{r \in \cup_{a \in X} Q_a} c_r$, excluding previously paid resources.
- (D) The action pool and signed map are fixed and known. Their construction and information-access costs are separately disclosed.

Condition (A) is sufficient and deliberately strong. Under independently selectable actions, (B) also has a direct polynomial check.

Lemma 5.1 (Checking unique activation). *For a reachable witness, let M_w be the set of actions that are sole providers of some uncovered required atom. Its unique minimal activating set exists exactly when M_w covers all remaining uncovered required atoms; then $D_w = M_w$.*

Proof. Each sole provider belongs to every activating set. Conversely, if an action is not a sole provider, selecting all other actions still covers every required atom not already in E_0 , so that action is not mandatory. If the finite family of activating sets has a unique inclusion-minimal member, every activating set contains it: repeated deletion reaches a minimal member. That member therefore consists exactly of the mandatory actions and must cover all remaining premises. Conversely, if M_w covers those premises, together with E_0 it activates the witness and is contained in every activating set, making it uniquely minimal. $\square$

For every distinct nonempty useful D_w , create a macro j with primitive actions D_w , cost $c_j = c(D_w)$, and obligation coverage Q_j obtained by evaluating that union. Remove empty-coverage macros and duplicates. Let ρ be the maximum number of retained macros containing a positive-cost resource; set $\rho = 1$ if there are no positive-cost resources. These macros are plans to purchase complete premises, not free summaries.

Apply weighted greedy set cover to the fixed (Q_j, c_j) : take useful zero-cost macros first, otherwise minimize cost per newly covered obligation. For this analysis use the *full* macro cost, not an overlap-discounted marginal cost. Finally purchase the union of the chosen primitive actions.

Proposition 5.2 (Conditional cover bound). *Under (A)–(D), for a feasible target with $m \geq 1$, greedy returns a selection satisfying the support, refutation, and conflict constraints, of incremental cost at most $\rho H_m \text{OPT}_h$, where $H_m = \sum_{i=1}^m 1/i$. Returning the cheaper of this result and the union of the cheapest macro for each obligation gives*

$$c(X) \leq \min\{m, \rho H_m\} \text{OPT}_h.$$

For $m = 0$, return the initial evidence without additional actions.

Proof. Every obligation satisfied by an optimal primitive selection X^* has an active witness; (B) gives a macro $D_w \subseteq X^*$. These macros cover O_h , and each resource of X^* is charged in at most ρ macros. Thus an additive macro cover costs at most $\rho c(X^*)$.

For completeness, the standard weighted greedy argument (Chvátal, 1979) is as follows. With r obligations uncovered, an optimal additive cover of cost T has total residual coverage at least r ; some set has cost per newly covered obligation at most T/r . Charge each newly covered obligation the chosen ratio. The j th obligation covered receives charge at most $T/(m - j + 1)$, so total cost is at most $H_m T$. The actual resource union costs no more than this sum of macro costs.

Alternatively, for each obligation its witness macro inside X^* costs at most $c(X^*)$; the cheapest eligible macro costs no more. Summing over m obligations proves the m bound. Either union covers every obligation and is conflict-free by (A), hence satisfies the stated structural constraints. $\square$

This is a classical cover corollary with explicit union-cost comparison, not a new approximation theorem. Disjoint macro charge resources give $\rho = 1$. If (A)–(D) hold for every feasible target, choosing the cheapest verified result gives the corresponding uniform bound against the global optimum. Otherwise its comparator is restricted to eligible targets.

The coverage gate and actual budget must still pass. This approximation is not budget-preserving: its output may exceed B even when the optimum fits. The guarantee is single-objective and has no extra hard side constraints. For example, minimizing acquisition cost with a separate exact rendered-token capacity is not covered: check that capacity and use the exact constrained solver where needed. Failure of (A) can make a macro union conflict. Failure of (B) requires enumerating all minimal premise-covering action sets, possibly exponentially many; a shortlist supports only a shortlist-optimum guarantee. Failure of (C) prevents calling the bound a native-token bound. Bounded premise arity alone does not restore submodularity. Every bound remains conditional on the supplied signed map and independent coverage assessment.

6 An operational acquisition policy

The exact certificate problem is directly useful after the signed witness map is available. It is not, by itself, an online retriever: obtaining that map may already require reading and verifying all candidate evidence. We specify the following implementable research policy, called *certificate-directed acquisition*, to make the intended empirical method concrete. Its planning predictions have no authority to certify an answer.

1. **Declare the task.** Parse all required slots, versions and relation types; retain alternative material query readings. Build typed complete joint proposals from observed source occurrences and permitted structural descriptors. Initialize the coverage sentinel to open. A changed question, candidate pruning, or an expanded source domain requires reassessing the coverage assertion rather than reusing it silently.
2. **Expose candidate actions.** Offer whole source neighborhoods such as a row with its possible caption, a continuation identity, or a definition footnote. Record their source identifiers, preview access, estimated cost and actual acquisition cost. Reading a preview is information access and is charged. Reference annotations are inaccessible to this stage.
3. **Construct prospective obligations.** For each possible target, list missing positive claims and each rival requiring refutation. A planner may predict which complete witness premises could discharge them. Use a bounded exact solution of this *predicted* map, or a declared heuristic, to propose a complete acquisition plan. Prefer plans for which all required dependencies can fit; this allows an initially zero-gain action that completes a later conjunction. Predictions and their uncertainty remain separate from the observed judgment ledger. While the sentinel is open, this is diagnostic optimization of the structural support and exclusion obligations only; a hypothetical plan does not clear the sentinel or authorize an answer.
4. **Acquire, then verify.** Acquire a whole action from the chosen plan and charge it immediately. Validate received anchors. Only after all required premises are present may a source-conditioned verifier add support or contradiction to the observed ledger. An unresolved judgment adds neither. Retain adverse results and conflicts; replan when a predicted witness fails. If no predicted plan is available, apply a frozen exploration rule or stop. This choice, tie breaking, shortlist size and compute limit are fixed on development data, not selected after held-out outcomes.
5. **Check and render.** Apply the certificate gate to observed judgments and the current coverage assessment, with an exact final rendered cost check. If it passes, render the complete typed

answer and its source derivations. Otherwise return insufficient evidence, a request to clarify the unresolved query distinction, or an explicitly scored partial answer. Stop at the declared acquisition or compute limit.

There are two ledgers. The acquisition ledger retains every source read, proposal and verifier call, including failed plans and evidence later omitted from the reader input. The bounded reader pack carries every premise used by the returned certificate. Shared spans are charged once within a union but are never free. Source acquisition, final prompt tokens and latency are reported in their own units; a scalar objective requires an explicit conversion. Known active contradictions cannot be hidden by dropping their spans from a later pack. A conservative implementation checks the full observed ledger for conflicts in addition to the selected pack. The v21 signed ledger therefore abstains on an observed conflict: adding another judgment cannot erase two already active signs. Adjudication or rule supersession would need a separately versioned, justified protocol that preserves the original judgments, outside the present guarantee.

Proposition 6.1 (Policy-independent conditional acceptance). *If every emitted answer satisfies the sound certificate contract, adaptive planning, predicted-witness ranking and stopping do not change deterministic source-relative soundness. They may change answer coverage and cost.*

Proof. Apply the source-relative soundness theorem to the final observed certificate on each emitted run. Its premises concern the final evidence, actual judgments, coverage assertion and renderer, not the path that proposed an action. A prediction affects the path only. It cannot replace a premise of that theorem. For a stochastic learned verifier, satisfying its error assumptions after adaptive selection remains necessary; the selective-risk analysis already accounts for that distinction. $\square$

This policy is a specified research method, not a completed automatic pipeline in the current repository. There is no cost-competitive online theorem for it. In particular, a surrogate plan may be wrong, a source may not contain a needed witness, and a finite proposal beam may miss a valid answer class. The exact reference solver diagnoses finite packing; it cannot remove these semantic and discovery errors.

7 Relationship to prior work

Decision classes and adaptive tests. EC² already selects costly tests to distinguish equivalence classes, rather than every latent hypothesis (Golovin et al., 2010). Its formulation can represent correlated noise through expanded realizations. ECED uses a root-cause model with specified test likelihoods (Chen et al., 2017). A learned score over source-binding proposals does not by itself supply these probability models. Our joint answer map specializes the class definition; it is not a new equivalence-class objective.

The expected-cover proof underlying the original adaptive-submodularity analysis was corrected in the authors’ version 5, with a weaker bound under additional assumptions (Golovin and Krause, 2011). Later cover results have their own models and parameters (Esfandiari et al., 2021; Hellerstein et al., 2021). We do not transfer the original EC² constant to this problem or assert that EC² itself has been disproved. More fundamentally, conjunctive source rules need not have the diminishing returns required by the relevant guarantees.

Logical certificates and finite optimization. Test cover reduces relevant distinctions to a pair universe (de Bontridder et al., 2003). Abduction-based explanations combine consistency and entailment and can be computed with exact hitting-set methods (Ignatiev et al., 2019a); explanation/counterexample duality is also established (Ignatiev et al., 2019b). Boolean-function evaluation

separates a certificate’s cost from the adaptive cost of finding it (Deshpande et al., 2014). These precedents rule out claims that minimum certificates, positive evidence plus exclusion, or offline/online distinctions are new general principles. Our finite encoding specifies how source anchors, complete joint answers, conjunctive premises and conflicts enter that contract.

Sufficiency and complementary evidence. Sufficient Context considers ambiguity and selective answering (Joren et al., 2025). S2G-RAG’s structured gaps include relations and evidence spans; its sentence-pointer mechanism retains disambiguating context (Li et al., 2026). RINSE evaluates evidence adequacy before answer generation (Chen et al., 2026). Therefore a generic sufficiency baseline can request the same table, definition or scope witness as our proposed controller.

The Missing Complement explicitly represents alternatives within evidence requirements and conjunction across requirements, and distinguishes selection from source discovery (Feng et al., 2026). Its coding-agent setting differs from source-scoped factual answers, but conjunctive evidence recovery and set-level complement acquisition are direct prior mechanisms. The intended empirical distinction here is whether explicit competing joint answers and verified rival exclusion improve the allocation of the same evidence and controller budget.

Ambiguous and conflicting answers. Tree of Clarifications, REFUEL and MADAM-RAG already explore interpretations or resolve conflicting evidence (Kim et al., 2023; Gao et al., 2021; Wang et al., 2025). Candidate generation, pruning and disagreement alone cannot carry our novelty claim. Their mechanisms should inform matched ambiguity-aware baselines, including cases where clarification is the appropriate action.

Unresolved related-work scope. The indexed primary abstract of ROSE-RAG describes robust sequential evidence acquisition under source-model ambiguity (Mokhtari and Davoodi, 2026). Its full theorem assumptions and proofs were unavailable in this audit. We therefore record a potential direct overlap rather than claim a verified distinction. The audit is current to 5 October 2026 and is targeted, not an exhaustive novelty certification. It must be refreshed before submission.

Result in this manuscript	Contribution status
Joint scoped certificate	Domain-specific integration and explicit semantic assurance contract; novelty of the integration remains to be defended.
Soundness and error bounds	Elementary consequences of stated premises and event containment; not new concentration or calibration theory.
Exact finite optimization	Boolean/MILP encoding and reference enumeration; not a new general solver.
Hardness and cover bound	Set-cover reduction and attributed conditional adaptation; no unrestricted greedy guarantee.
Lookahead counterexample	A precise limitation of the specified policy class; no lower bound against all acquisition algorithms.
Retrieval improvement	Prospective empirical claim; no supporting held-out result in this checkpoint.

Table 1: Claim boundaries that the empirical paper must preserve.

8 Empirical contract and falsifiable claims

The local study must test the semantic premises and practical value separately. Use immutable source snapshots and source-family-disjoint development and test splits. Create references before method outputs with independent source review and adjudication. Familiar histories and authored

mathematical controls remain development material. Multiple variants of one history are dependent samples.

The primary endpoint should be the rate of complete, jointly correct and source-supported answers under a prespecified final-input budget and charged controller budget. Also report answer coverage, conditional error among emitted answers, wrong emitted answers per all questions, clarification and abstention, and every timeout or schema failure. For a question with genuinely multiple admissible readings, the reference must permit clarification or a complete scoped multi-answer contract; an arbitrary single reference cannot establish unique resolution.

Measure candidate coverage on valid representatives, false support, false elimination of valid rivals, premise completeness, output faithfulness and actual cost. Mere action-label recall does not validate candidate coverage. Claimwise verifier accuracy does not replace runwise or adaptively selected error analysis. The theoretical error terms are population quantities; no numerical bound follows until those quantities are estimated on appropriate independent data.

The baseline ladder is: a strong reader with structure-preserving closure; typed scope/version filtering; relation-aware coverage selection; a strong reranker; a gap/sufficiency controller; and an ambiguity-aware controller. Give every method the same candidate source pool, available structure, reader, abstention policy and charged access budget. Add compute-matched semantic reranking and set-level complementary selection. Separate a supplied-pool packing diagnostic from source-only discovery. Compare the proposed policy to the closest available mechanisms, not only to a word-slot coverage strawman.

Fix the primary comparison, practically meaningful effect, sample size method and decision rule before held-out predictions. Use a separate pilot to estimate history-level variation and justify power; a recruitment target alone is not a power calculation. Use history/source-family clustered uncertainty for paired method differences. Report the tradeoff across budgets and answer coverage, not a selected best operating point. Preserve failed runs and all amendments.

The paper’s positive method claim requires a useful improvement over the strongest matched baseline with uncertainty that supports the prespecified effect, without concealing unsupported answers or extra compute. If structural closure or generic sufficiency solves the same cases at the same cost, the specialized method claim must be narrowed or abandoned. A formal certificate and successful authored controls alone do not meet this gate.

9 Artifact scope and limitations

The released versioned module `certificate_v21.py` evaluates a supplied atomic witness circuit and searches its finite action subsets. It is not a source extractor, natural-language entailment model, candidate-coverage oracle, or completed local model host. Atomic coverage must be derived and validated upstream; identifiers do not authenticate source meaning. The MILP and restricted approximation are mathematical specifications; their implementation status is listed separately from the executable exhaustive solver.

The accompanying `local_empirical_contract_v21.txt` lists commands, required adapters, metrics and launch gates. The proof review is a separate model-role review, not independent human peer review. Automated checks use authored finite cases and establish implementation behavior only. No native reader or natural QA experiment is run for this theory checkpoint. Earlier negative reader and retrieval results remain in the repository unchanged.

The main mathematical limitation is semantic: a finite checker cannot prove that unrestricted natural-language proposals cover every admissible class, that source assertions entail a supplied judgment, or that a source is true in the world. The exact optimizer is exponential. The restricted approximation requires a complete macro representation, conflict freedom and a declared additive

resource model. Exact tokenizer cost does not generally satisfy that model. The online policy has no general approximation guarantee. These limitations define measurable failure modes for the local study rather than unreported exceptions to a broad guarantee.

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